Supplementary Material for Audio Codec Effects on Feature Attribution in Speech Emotion Recognition

Akshath R

Section, equation and figure numbers without an S prefix refer to the main paper.

Appendix A
Complete Feature Inventory

Every stimulus is reduced to 436 named columns. Frame-level contours are summarised by eight order statistics (mean, standard deviation, minimum, maximum, median, interquartile range, skewness, excess kurtosis). Descriptors marked with a shorter list are summarised only by those statistics, and scalars are already clip-level. Naming is <family>_<descriptor>_<statistic> throughout, which is what makes an attribution over a column a claim about an acoustic quantity.

Analysis frames are 512 point FFT, 400 sample window, 160 sample hop at 16 kHz, which gives 25 ms windows every 10 ms. Praat measures use a 75 Hz to 500 Hz pitch range.

A. Descriptors and their definitions

TABLE I
Spectral descriptors. 114 columns of 436. "all 8" denotes the eight order statistics of Section III-C: mean, standard deviation, minimum, maximum, median, IQR, skewness and excess kurtosis.

Descriptor	Definition	Stats	n
Spectral centroid	First moment of the power spectrum. It is the frequency about which spectral energy is balanced. Correlates with perceived brightness.	all 8	8
Spectral bandwidth	Power-weighted second moment about the centroid. It measures how widely energy is spread in frequency.	all 8	8
Spectral roll-off (85 %)	Frequency below which 85 % of the frame's spectral energy lies.	all 8	8
Spectral roll-off (95 %)	As above at 95 %. Directly sensitive to codec low-pass behaviour.	all 8	8
Spectral flatness	Geometric mean of the magnitude spectrum divided by its arithmetic mean. It approaches 1 for noise-like frames and 0 for tonal ones, and it collapses when an encoder sets masked bins to zero.	all 8	8
Zero-crossing rate	Fraction of adjacent sample pairs with a sign change. It is a cheap proxy for voicing and frication.	all 8	8
Frame RMS energy	Root-mean-square magnitude per frame, which gives the loudness contour.	all 8	8
Spectral flux	Euclidean norm of the frame-to-frame difference of the magnitude spectrum. It measures how fast the spectrum is changing.	all 8	8
Spectral entropy	Shannon entropy of the power spectrum normalised to its own bin count, $-\sum_k P_k \ln P_k / \ln K$ with P_k the share of frame power in bin k of K. It is a scale-free measure of how evenly energy is distributed across frequency.	all 8	8
Spectral slope	Least-squares slope of magnitude regressed on frequency, per frame. This is the spectral tilt.	all 8	8
Band-energy ratios (8 bands)	Fraction of frame energy in each of 0–500, 500–1000, 1000–2000, 2000–3000, 3000–4000, 4000–5000, 5000–6000, 6000–8000 Hz. Energy-normalised, so these describe spectral shape rather than level.	mean, std	16
High-frequency survival ratios	Fraction of frame energy above 4 and above 6 kHz. The most direct probes of lossy-codec low-pass behaviour in the inventory.	mean, std	4
Spectral contrast (7 sub-bands)	Per-band difference between spectral peaks and valleys in decibels. Band 6 covers 6–8 kHz and is the single most codec-displaced column in the table (Section III-D).	mean, std	14

TABLE II
Cepstral descriptors. 240 columns of 436. "all 8" denotes the eight order statistics of Section III-C: mean, standard deviation, minimum, maximum, median, IQR, skewness and excess kurtosis.

Descriptor	Definition	Stats	n
ΔMFCC (20)	First-order temporal derivative of each MFCC, over a 9-frame window.	mean, std	40
ΔΔMFCC (20)	Second-order temporal derivative of each MFCC.	mean, std	40
MFCC (20)	Discrete cosine transform of the log mel-filterbank energies, which is the standard SER front end. MFCC-0 tracks overall log energy, so its derivatives measure how fast loudness changes.	all 8	160

TABLE III
Prosodic and voice-quality descriptors (Praat). 56 columns of 436. "all 8" denotes the eight order statistics of Section III-C: mean, standard deviation, minimum, maximum, median, IQR, skewness and excess kurtosis.

Descriptor	Definition	Stats	n
F_0 slope (mean $\lvert\cdot\rvert$)	Mean absolute first difference of the voiced F_0 contour, per second. It is the rate of pitch movement.	scalar	1
Fundamental frequency F_0	Praat autocorrelation pitch over voiced frames only.	all 8	8
Jitter (4 variants)	Cycle-to-cycle variation in glottal period: local, RAP, PPQ5, DDP. A perturbation measure of the voice source.	scalar	4
Shimmer (6 variants)	Cycle-to-cycle variation in glottal amplitude: local, local dB, APQ3, APQ5, APQ11, DDA. APQ11 needs eleven consecutive periods and is the one column in the table with non-trivial missingness (11.3 %).	scalar	6
Harmonics-to-noise ratio	Ratio of periodic to aperiodic energy in decibels. It correlates with hoarseness and breathiness.	mean, std, min, max, median	5
Formant bandwidths B_1–B_3	Median -3 dB bandwidth of the first three formants (Burg estimator).	median	1
Formants F_1–F_5	Burg-estimator formant frequencies, which describe the vocal-tract filter. F_2 movement is the strongest single correlate of disgust in this corpus.	mean, std, median	17
Intensity	Praat short-term intensity contour in decibels. Its standard deviation is among the top-ranked features for every model family in the study.	all 8	8
Voiced fraction	Proportion of frames with a defined F_0.	scalar	1
Pause fraction	Complement of the voiced fraction. By definition it equals 1− voiced fraction ($r = -1.00$).	scalar	1
Voiced-segment count	Number of maximal runs of voiced frames.	scalar	1
Voiced-segment rate	Voiced segments per second, which is a proxy for speech rate.	scalar	1
Mean voiced-segment duration	Mean length in seconds of a voiced run.	scalar	1
CPPS	Smoothed cepstral peak prominence, which is the standard acoustic correlate of breathy phonation. Drives both sadness and neutrality in every attribution method applied here.	scalar	1

TABLE IV
Chroma descriptors. 24 columns of 436. "all 8" denotes the eight order statistics of Section III-C: mean, standard deviation, minimum, maximum, median, IQR, skewness and excess kurtosis.

Descriptor	Definition	Stats	n
Chroma (12 pitch classes)	Energy folded onto the twelve semitone pitch classes. Tuning is pinned to 0 rather than estimated, because an estimated tuning could itself shift under compression and become part of the effect being measured.	mean, std	24

TABLE V
Global descriptors. 2 columns of 436. "all 8" denotes the eight order statistics of Section III-C: mean, standard deviation, minimum, maximum, median, IQR, skewness and excess kurtosis.

Descriptor	Definition	Stats	n
Clip duration	Decoded duration in seconds. Rises by 32.68 ms under AAC encoder priming, uniformly across every clip.	scalar	1
Peak amplitude	Maximum absolute sample value after loudness normalisation.	scalar	1

B. Enumerated column names

The complete inventory, in the order the extractor emits it.

TABLE VI
Spectral descriptors: all 114 column names.

spectral__centroid__mean	spectral__rolloff95__iqr	spectral__flux__min	spectral__band__2000__3000__std
spectral__centroid__std	spectral__rolloff95__skew	spectral__flux__max	spectral__band__3000__4000__mean
spectral__centroid__min	spectral__rolloff95__kurt	spectral__flux__median	spectral__band__3000__4000__std
spectral__centroid__max	spectral__flatness__mean	spectral__flux__iqr	spectral__band__4000__5000__mean
spectral__centroid__median	spectral__flatness__std	spectral__flux__skew	spectral__band__4000__5000__std
spectral__centroid__iqr	spectral__flatness__min	spectral__flux__kurt	spectral__band__5000__6000__mean
spectral__centroid__skew	spectral__flatness__max	spectral__entropy__mean	spectral__band__5000__6000__std
spectral__centroid__kurt	spectral__flatness__median	spectral__entropy__std	spectral__band__6000__8000__mean
spectral__bandwidth__mean	spectral__flatness__iqr	spectral__entropy__min	spectral__band__6000__8000__std
spectral__bandwidth__std	spectral__flatness__skew	spectral__entropy__max	spectral__hf__ratio__4000__mean
spectral__bandwidth__min	spectral__flatness__kurt	spectral__entropy__median	spectral__hf__ratio__4000__std
spectral__bandwidth__max	spectral__zcr__mean	spectral__entropy__iqr	spectral__hf__ratio__6000__mean
spectral__bandwidth__median	spectral__zcr__std	spectral__entropy__skew	spectral__hf__ratio__6000__std
spectral__bandwidth__iqr	spectral__zcr__min	spectral__entropy__kurt	spectral__contrast__0__mean
spectral__bandwidth__skew	spectral__zcr__max	spectral__slope__mean	spectral__contrast__0__std
spectral__bandwidth__kurt	spectral__zcr__median	spectral__slope__std	spectral__contrast__1__mean
spectral__rolloff85__mean	spectral__zcr__iqr	spectral__slope__min	spectral__contrast__1__std
spectral__rolloff85__std	spectral__zcr__skew	spectral__slope__max	spectral__contrast__2__mean
spectral__rolloff85__min	spectral__zcr__kurt	spectral__slope__median	spectral__contrast__2__std
spectral__rolloff85__max	spectral__rms__mean	spectral__slope__iqr	spectral__contrast__3__mean
spectral__rolloff85__median	spectral__rms__std	spectral__slope__skew	spectral__contrast__3__std
spectral__rolloff85__iqr	spectral__rms__min	spectral__slope__kurt	spectral__contrast__4__mean
spectral__rolloff85__skew	spectral__rms__max	spectral__band__0__500__mean	spectral__contrast__4__std
spectral__rolloff85__kurt	spectral__rms__median	spectral__band__0__500__std	spectral__contrast__5__mean
spectral__rolloff95__mean	spectral__rms__iqr	spectral__band__500__1000__mean	spectral__contrast__5__std
spectral__rolloff95__std	spectral__rms__skew	spectral__band__500__1000__std	spectral__contrast__6__mean
spectral__rolloff95__min	spectral__rms__kurt	spectral__band__1000__2000__mean	spectral__contrast__6__std
spectral__rolloff95__max	spectral__flux__mean	spectral__band__1000__2000__std	
spectral__rolloff95__median	spectral__flux__std	spectral__band__2000__3000__mean	

TABLE VII
Cepstral descriptors: all 240 column names.

mfcc__00__mean	mfcc__07__median	mfcc__15__mean	mfcc__d1__05__mean
mfcc__00__std	mfcc__07__iqr	mfcc__15__std	mfcc__d1__05__std
mfcc__00__min	mfcc__07__skew	mfcc__15__min	mfcc__d2__05__mean
mfcc__00__max	mfcc__07__kurt	mfcc__15__max	mfcc__d2__05__std
mfcc__00__median	mfcc__08__mean	mfcc__15__median	mfcc__d1__06__mean
mfcc__00__iqr	mfcc__08__std	mfcc__15__iqr	mfcc__d1__06__std
mfcc__00__skew	mfcc__08__min	mfcc__15__skew	mfcc__d2__06__mean
mfcc__00__kurt	mfcc__08__max	mfcc__15__kurt	mfcc__d2__06__std
mfcc__01__mean	mfcc__08__median	mfcc__16__mean	mfcc__d1__07__mean
mfcc__01__std	mfcc__08__iqr	mfcc__16__std	mfcc__d1__07__std
mfcc__01__min	mfcc__08__skew	mfcc__16__min	mfcc__d2__07__mean
mfcc__01__max	mfcc__08__kurt	mfcc__16__max	mfcc__d2__07__std
mfcc__01__median	mfcc__09__mean	mfcc__16__median	mfcc__d1__08__mean
mfcc__01__iqr	mfcc__09__std	mfcc__16__iqr	mfcc__d1__08__std
mfcc__01__skew	mfcc__09__min	mfcc__16__skew	mfcc__d2__08__mean
mfcc__01__kurt	mfcc__09__max	mfcc__16__kurt	mfcc__d2__08__std
mfcc__02__mean	mfcc__09__median	mfcc__17__mean	mfcc__d1__09__mean
mfcc__02__std	mfcc__09__iqr	mfcc__17__std	mfcc__d1__09__std
mfcc__02__min	mfcc__09__skew	mfcc__17__min	mfcc__d2__09__mean
mfcc__02__max	mfcc__09__kurt	mfcc__17__max	mfcc__d2__09__std
mfcc__02__median	mfcc__10__mean	mfcc__17__median	mfcc__d1__10__mean
mfcc__02__iqr	mfcc__10__std	mfcc__17__iqr	mfcc__d1__10__std
mfcc__02__skew	mfcc__10__min	mfcc__17__skew	mfcc__d2__10__mean
mfcc__02__kurt	mfcc__10__max	mfcc__17__kurt	mfcc__d2__10__std
mfcc__03__mean	mfcc__10__median	mfcc__18__mean	mfcc__d1__11__mean
mfcc__03__std	mfcc__10__iqr	mfcc__18__std	mfcc__d1__11__std
mfcc__03__min	mfcc__10__skew	mfcc__18__min	mfcc__d2__11__mean
mfcc__03__max	mfcc__10__kurt	mfcc__18__max	mfcc__d2__11__std
mfcc__03__median	mfcc__11__mean	mfcc__18__median	mfcc__d1__12__mean
mfcc__03__iqr	mfcc__11__std	mfcc__18__iqr	mfcc__d1__12__std
mfcc__03__skew	mfcc__11__min	mfcc__18__skew	mfcc__d2__12__mean
mfcc__03__kurt	mfcc__11__max	mfcc__18__kurt	mfcc__d2__12__std
mfcc__04__mean	mfcc__11__median	mfcc__19__mean	mfcc__d1__13__mean
mfcc__04__std	mfcc__11__iqr	mfcc__19__std	mfcc__d1__13__std
mfcc__04__min	mfcc__11__skew	mfcc__19__min	mfcc__d2__13__mean
mfcc__04__max	mfcc__11__kurt	mfcc__19__max	mfcc__d2__13__std
mfcc__04__median	mfcc__12__mean	mfcc__19__median	mfcc__d1__14__mean
mfcc__04__iqr	mfcc__12__std	mfcc__19__iqr	mfcc__d1__14__std
mfcc__04__skew	mfcc__12__min	mfcc__19__skew	mfcc__d2__14__mean
mfcc__04__kurt	mfcc__12__max	mfcc__19__kurt	mfcc__d2__14__std
mfcc__05__mean	mfcc__12__median	mfcc__d1__00__mean	mfcc__d1__15__mean
mfcc__05__std	mfcc__12__iqr	mfcc__d1__00__std	mfcc__d1__15__std
mfcc__05__min	mfcc__12__skew	mfcc__d2__00__mean	mfcc__d2__15__mean
mfcc__05__max	mfcc__12__kurt	mfcc__d2__00__std	mfcc__d2__15__std
mfcc__05__median	mfcc__13__mean	mfcc__d1__01__mean	mfcc__d1__16__mean
mfcc__05__iqr	mfcc__13__std	mfcc__d1__01__std	mfcc__d1__16__std
mfcc__05__skew	mfcc__13__min	mfcc__d2__01__mean	mfcc__d2__16__mean
mfcc__05__kurt	mfcc__13__max	mfcc__d2__01__std	mfcc__d2__16__std
mfcc__06__mean	mfcc__13__median	mfcc__d1__02__mean	mfcc__d1__17__mean
mfcc__06__std	mfcc__13__iqr	mfcc__d1__02__std	mfcc__d1__17__std
mfcc__06__min	mfcc__13__skew	mfcc__d2__02__mean	mfcc__d2__17__mean
mfcc__06__max	mfcc__13__kurt	mfcc__d2__02__std	mfcc__d2__17__std
mfcc__06__median	mfcc__14__mean	mfcc__d1__03__mean	mfcc__d1__18__mean
mfcc__06__iqr	mfcc__14__std	mfcc__d1__03__std	mfcc__d1__18__std
mfcc__06__skew	mfcc__14__min	mfcc__d2__03__mean	mfcc__d2__18__mean
mfcc__06__kurt	mfcc__14__max	mfcc__d2__03__std	mfcc__d2__18__std
mfcc__07__mean	mfcc__14__median	mfcc__d1__04__mean	mfcc__d1__19__mean
mfcc__07__std	mfcc__14__iqr	mfcc__d1__04__std	mfcc__d1__19__std
mfcc__07__min	mfcc__14__skew	mfcc__d2__04__mean	mfcc__d2__19__mean
mfcc__07__max	mfcc__14__kurt	mfcc__d2__04__std	mfcc__d2__19__std

TABLE VIII
Prosodic and voice-quality descriptors (Praat): all 56 column names.

prosody__f0__mean	prosody__jitter__local	prosody__hnr__median	prosody__f4__std
prosody__f0__std	prosody__jitter__rap	prosody__f1__mean	prosody__f4__median
prosody__f0__min	prosody__jitter__ppq5	prosody__f1__std	prosody__f5__mean
prosody__f0__max	prosody__jitter__ddp	prosody__f1__median	prosody__f5__std
prosody__f0__median	prosody__shimmer__local	prosody__f1__bw__median	prosody__f5__median
prosody__f0__iqr	prosody__shimmer__localdb	prosody__f2__mean	prosody__intensity__mean
prosody__f0__skew	prosody__shimmer__apq3	prosody__f2__std	prosody__intensity__std
prosody__f0__kurt	prosody__shimmer__apq5	prosody__f2__median	prosody__intensity__min
prosody__voiced__fraction	prosody__shimmer__apq11	prosody__f2__bw__median	prosody__intensity__max
prosody__f0__slope__abs__mean	prosody__shimmer__dda	prosody__f3__mean	prosody__intensity__median
prosody__n__voiced__segments	prosody__hnr__mean	prosody__f3__std	prosody__intensity__iqr
prosody__voiced__rate__per__s	prosody__hnr__std	prosody__f3__median	prosody__intensity__skew
prosody__mean__voiced__seg__s	prosody__hnr__min	prosody__f3__bw__median	prosody__intensity__kurt
prosody__pause__fraction	prosody__hnr__max	prosody__f4__mean	prosody__cpps

TABLE IX
Chroma descriptors: all 24 column names.

chroma__00__mean	chroma__04__mean	chroma__08__mean
chroma__00__std	chroma__04__std	chroma__08__std
chroma__01__mean	chroma__05__mean	chroma__09__mean
chroma__01__std	chroma__05__std	chroma__09__std
chroma__02__mean	chroma__06__mean	chroma__10__mean
chroma__02__std	chroma__06__std	chroma__10__std
chroma__03__mean	chroma__07__mean	chroma__11__mean
chroma__03__std	chroma__07__std	chroma__11__std

TABLE X
Global descriptors: all 2 column names.

duration__s	peak__amplitude

Appendix B
Per-Condition Tree Splits

The tables below list every internal node down to depth 4 of the entropy tree of Section V-E. The tree was fitted independently on each condition, using the same actors, the same items and the same hyper-parameters. The Path column reads from the root, and L denotes the branch taken when the parent's test is satisfied. Entropies are given in bits. The value n is the raw sample count, and the information gain is computed on the class-weighted counts (Eq. 3).

A path missing from a table is one the tree stopped splitting, because a child would have fallen below the 25-sample leaf minimum. Comparing the same path across the four tables is the node-level comparison summarised in Fig. 4 of the main paper.

TABLE XI
Entropy tree fitted on Uncompressed WAV, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody_intensity_std	8.2184	2.5850	0.3090	4905
L	mfcc_d1_00_std	3.9965	2.3849	0.1415	3065
R	mfcc_d1_00_std	6.6630	2.0891	0.1797	1840
LL	spectral_band_0_500_mean	0.8626	1.9382	0.0685	1059
LR	prosody_f0_mean	220.5886	2.3995	0.0990	2006
RL	prosody_f0_median	161.8673	2.2633	0.1141	847
RR	prosody_intensity_min	42.7554	1.6055	0.0925	993
LLL	prosody_intensity_median	64.3901	2.1110	0.0763	503
LLR	prosody_f0_mean	174.4501	1.6481	0.0727	556
LRL	spectral_centroid_iqr	209.5395	2.3328	0.0682	1781
LRR	prosody_f0_max	372.4767	2.0340	0.1797	225
RLL	spectral_centroid_iqr	310.5503	2.3335	0.1579	254
RLR	prosody_f0_min	149.0330	2.0683	0.0778	593
RRL	mfcc_03_max	46.1135	1.2265	0.0747	542
RRR	spectral_contrast_2_mean	12.1160	1.8550	0.0915	451
LLLL	mfcc_05_iqr	8.7058	2.3273	0.1863	139
LLLR	mfcc_07_mean	1.0004	1.9225	0.0759	364
LLRL	mfcc_00_mean	-220.3358	1.6344	0.0676	391
LLRR	prosody_intensity_iqr	5.8890	1.4344	0.1343	165
LRLL	mfcc_d1_03_std	1.3160	2.2210	0.0810	564
LRLR	duration_s	2.6526	2.2848	0.0690	1217
LRRL	mfcc_d1_00_std	5.5015	2.2359	0.2615	119
LRRR	mfcc_d2_05_std	0.6661	1.4206	0.2061	106
RLLL	spectral_contrast_1_std	3.1735	2.2781	0.1619	146
RLLR	duration_s	2.4524	2.0339	0.1987	108
RLRL	mfcc_03_std	12.0396	2.1028	0.0993	385
RLRR	prosody_intensity_std	10.4187	1.7818	0.1437	208
RRLL	spectral_entropy_mean	0.4682	1.5107	0.1241	305
RRLR	mfcc_d1_00_std	8.0503	0.6898	0.0951	237
RRRL	mfcc_d2_00_std	3.2530	2.1332	0.2152	95
RRRR	mfcc_03_max	40.8027	1.6633	0.1040	356

TABLE XII
Entropy tree fitted on MP3 @ 64 kbit/s, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody_intensity_std	8.2210	2.5850	0.3090	4905
L	mfcc_d1_00_std	4.0175	2.3849	0.1398	3065
R	mfcc_d1_00_std	6.6883	2.0891	0.1800	1840
LL	spectral_band_0_500_mean	0.8627	1.9524	0.0669	1080
LR	prosody_f0_mean	221.5304	2.3997	0.1002	1985
RL	prosody_f0_median	161.8687	2.2615	0.1129	873
RR	prosody_intensity_min	42.3291	1.5890	0.0914	967
LLL	prosody_intensity_median	63.9287	2.1257	0.0752	515
LLR	prosody_f0_mean	175.0993	1.6634	0.0762	565
LRL	spectral_band_0_500_std	0.2524	2.3332	0.0688	1771
LRR	mfcc_02_std	14.7187	2.0072	0.1668	214
RLL	spectral_centroid_iqr	310.5299	2.3306	0.1592	257
RLR	spectral_band_0_500_mean	0.7316	2.0707	0.0767	616
RRL	spectral_band_3000_4000_mean	0.0146	1.2123	0.0774	532
RRR	spectral_contrast_2_mean	12.2081	1.8440	0.1046	435
LLLL	mfcc_05_iqr	8.7206	2.3261	0.1391	148
LLLR	spectral_flatness_mean	0.0041	1.9388	0.0744	367
LLRL	mfcc_00_mean	-223.6980	1.6513	0.0753	399
LLRR	prosody_intensity_iqr	5.8541	1.4319	0.1347	166
LRLL	prosody_hnr_median	3.7622	2.2810	0.0703	935
LRLR	mfcc_d2_00_std	2.4823	2.2461	0.0777	836
LRRL	mfcc_06_mean	-1.1106	1.4472	0.3107	65
LRRR	mfcc_16_std	4.4168	2.0110	0.1969	149
RLLL	mfcc_05_median	2.2367	2.2728	0.1562	147
RLLR	duration_s	2.4524	2.0326	0.2006	110
RLRL	prosody_f0_median	270.2333	2.0442	0.1015	320
RLRR	prosody_f0_std	46.8924	1.9399	0.1148	296
RRLL	mfcc_03_max	39.2929	1.4878	0.1099	272
RRLR	mfcc_d2_10_std	0.6478	0.7657	0.0986	260
RRRL	mfcc_d2_00_std	3.2140	2.1218	0.2554	106
RRRR	mfcc_d2_12_std	0.5605	1.6136	0.0943	329

TABLE XIII
Entropy tree fitted on MP4/AAC @ 64 kbit/s, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody_intensity_std	8.3241	2.5850	0.3064	4905
L	mfcc_00_iqr	60.5853	2.3887	0.1408	3075
R	mfcc_d1_00_std	7.1056	2.0879	0.1555	1830
LL	spectral_band_500_1000_mean	0.0797	1.9819	0.0875	1181
LR	prosody_f0_mean	219.0372	2.4096	0.0957	1894
RL	prosody_f0_median	161.0991	2.2673	0.1047	874
RR	prosody_intensity_std	11.2496	1.6245	0.0884	956
LLL	prosody_f0_mean	169.4067	1.4798	0.0877	395
LLR	prosody_f3_bw_median	712.6446	2.0975	0.0757	786
LRL	spectral_band_0_500_mean	0.8284	2.3465	0.0696	1662
LRR	mfcc_02_std	14.7880	2.0711	0.1528	232
RLL	spectral_band_1000_2000_std	0.0765	2.3608	0.1810	226
RLR	spectral_band_0_500_mean	0.7133	2.0917	0.0718	648
RRL	spectral_contrast_2_mean	12.2765	1.8882	0.1074	461
RRR	mfcc_03_max	38.0092	1.2059	0.0851	495
LLLL	mfcc_05_iqr	7.6452	1.3054	0.1059	241
LLLR	prosody_f2_std	326.8660	1.5276	0.1362	154
LLRL	prosody_f0_mean	199.2747	2.1863	0.1157	498
LLRR	spectral_contrast_1_mean	7.7606	1.7295	0.0923	288
LRLL	duration_s	2.8480	2.2806	0.0690	1204
LRLR	prosody_cpps	5.3755	2.2671	0.0731	458
LRRL	mfcc_13_kurt	0.3511	1.5158	0.2506	66
LRRR	spectral_band_5000_6000_std	0.0311	2.0779	0.1904	166
RLLL	mfcc_00_iqr	112.3155	2.2882	0.1462	170
RLLR	spectral_band_500_1000_mean	0.1727	1.8385	0.2681	56
RLRL	prosody_f0_median	270.2040	2.0207	0.1057	296
RLRR	prosody_f0_slope_abs_mean	704.7569	2.0192	0.1053	352
RRLL	mfcc_05_median	3.1128	2.1051	0.1682	126
RRLR	spectral_band_0_500_std	0.2993	1.6568	0.1037	335
RRRL	prosody_f4_median	3950.6527	1.5464	0.1841	122
RRRR	spectral_band_3000_4000_mean	0.0143	0.9815	0.0752	373

TABLE XIV
Entropy tree fitted on Roundtrip WAV (AAC re-wrapped), all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody_intensity_std	8.3241	2.5850	0.3064	4905
L	mfcc_00_iqr	60.5802	2.3887	0.1408	3075
R	mfcc_d1_00_std	7.1058	2.0879	0.1555	1830
LL	spectral_band_500_1000_mean	0.0782	1.9819	0.0882	1181
LR	prosody_f0_mean	219.0372	2.4096	0.0957	1894
RL	prosody_f0_median	161.0993	2.2673	0.1047	874
RR	prosody_intensity_std	11.2497	1.6245	0.0884	956
LLL	prosody_f0_mean	170.6874	1.4582	0.0902	381
LLR	prosody_f3_bw_median	712.5267	2.0961	0.0727	800
LRL	spectral_band_0_500_mean	0.8280	2.3465	0.0701	1662
LRR	mfcc_02_std	14.7881	2.0711	0.1528	232
RLL	spectral_band_1000_2000_std	0.0765	2.3608	0.1810	226
RLR	spectral_band_0_500_mean	0.7136	2.0917	0.0728	648
RRL	spectral_contrast_2_mean	12.2826	1.8882	0.1051	461
RRR	mfcc_03_max	42.9480	1.2059	0.0846	495
LLLL	mfcc_02_min	-4.2720	1.3393	0.0935	239
LLLR	spectral_flatness_median	0.0075	1.4163	0.1184	142
LLRL	prosody_f0_mean	199.2748	2.1897	0.1143	505
LLRR	spectral_contrast_1_mean	7.7596	1.7311	0.0940	295
LRLL	duration_s	2.8480	2.2798	0.0680	1203
LRLR	prosody_f1_bw_median	281.4372	2.2671	0.0744	459
LRRL	mfcc_13_kurt	0.3489	1.5158	0.2506	66
LRRR	spectral_band_5000_6000_std	0.0311	2.0779	0.1904	166
RLLL	mfcc_00_iqr	112.3050	2.2882	0.1462	170
RLLR	spectral_band_500_1000_mean	0.1727	1.8385	0.2681	56
RLRL	prosody_f0_median	270.2047	2.0178	0.1036	299
RLRR	prosody_f0_slope_abs_mean	704.7769	2.0200	0.1026	349
RRLL	spectral_hf_ratio_4000_std	0.0443	2.1003	0.1608	128
RRLR	spectral_band_0_500_std	0.3006	1.6592	0.1006	333
RRRL	spectral_band_0_500_mean	0.6868	1.5345	0.1361	210
RRRR	spectral_band_3000_4000_mean	0.0144	0.8168	0.1042	285